

HENDRICKSON CREEK, AB, Canada

WMO: 719790

Lat: 53.795N

Lon: 118.447W

Elev: 1448

StdP: 85.10

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-35.6	-30.9	-39.1	0.1	-35.4	-34.1	0.2	-30.7	6.4	-0.4	6.1	-0.6	0.7	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	25.1	13.0	23.0	12.3	21.1	11.6	14.3	21.7	13.4	20.7	12.5	19.5	2.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.8	10.3	15.9	10.6	9.5	14.7	9.5	8.8	13.8	45.0	22.0	42.2	20.7	39.5	19.7	18.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.8	4.2	DB	-41.2	28.1	3.0	1.8	-43.3	29.5	-45.0	30.5	-46.7	31.6	-48.9	32.9	(4)
(5)			WB	-40.1	16.2	4.2	1.0	-43.2	16.9	-45.7	17.4	-48.0	18.0	-51.1	18.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-10.3	-9.2	-6.2	-0.7	4.8	8.6	11.3	10.5	6.4	0.6	-6.9	-11.2	(6)
(7)		DBStd	9.97	9.12	7.77	7.08	4.66	3.86	2.82	2.88	3.16	4.18	4.89	7.28	7.95	(7)
(8)		HDD10.0	3846	631	538	502	321	166	59	19	34	121	292	507	657	(8)
(9)		HDD18.3	6746	889	772	760	571	421	292	218	244	357	549	757	915	(9)
(10)		CDD10.0	142	0	0	0	0	3	17	59	48	14	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	154	0	0	0	0	3	16	62	51	21	1	0	0	(12)
(13)		CDH26.7	17	0	0	0	0	0	2	7	6	2	0	0	0	(13)
(14)	Wind	WSAvg	1.7	1.6	1.5	1.6	1.8	1.8	2.0	1.7	1.7	1.5	1.8	1.3	1.7	(14)
(15)	Precipitation	PrecAvg	642	44	29	37	33	58	88	88	70	52	49	51	44	(15)
(16)		PrecMax	806	87	53	63	64	110	142	178	110	98	80	102	74	(16)
(17)		PrecMin	492	9	10	10	15	31	27	42	36	19	25	17	21	(17)
(18)		PrecStd	68	19	11	13	12	17	27	33	21	19	15	18	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	9.2	12.9	18.0	23.2	25.7	27.5	27.5	26.0	19.7	10.8	7.4	(19)
(20)			MCWB	2.5	2.2	4.3	6.8	9.9	12.7	13.5	13.6	12.1	8.9	4.8	2.5	(20)
(21)		2%	DB	5.8	6.3	8.6	13.5	20.1	21.7	25.0	24.5	22.6	14.8	6.8	4.8	(21)
(22)			MCWB	1.3	1.0	2.3	5.3	8.9	11.1	13.1	13.3	11.6	7.3	2.3	0.6	(22)
(23)		5%	DB	3.9	3.8	6.2	10.7	17.6	19.7	23.0	22.2	19.8	11.5	4.4	2.2	(23)
(24)			MCWB	0.2	-0.2	0.9	3.6	8.0	10.1	12.6	12.6	10.5	5.6	0.9	-1.0	(24)
(25)		10%	DB	1.6	1.6	4.1	8.2	14.9	17.4	20.8	19.9	16.3	8.7	2.3	-0.2	(25)
(26)			MCWB	-1.2	-1.6	-0.4	2.2	6.8	9.3	12.1	12.0	9.4	4.0	-0.5	-2.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	3.2	4.8	7.6	10.9	13.6	15.8	15.5	13.9	9.9	4.9	2.9	(27)
(28)			MCD B	7.7	8.4	12.5	16.8	19.8	21.6	21.8	22.1	21.9	18.4	10.3	6.9	(28)
(29)		2%	WB	1.7	1.2	2.7	5.7	9.6	12.2	14.4	14.4	12.3	7.6	3.0	0.8	(29)
(30)			MCD B	5.3	5.3	7.8	12.6	18.2	19.7	21.9	21.5	21.0	13.5	6.3	4.2	(30)
(31)		5%	WB	0.4	-0.1	1.2	4.0	8.5	11.1	13.5	13.5	11.0	6.0	1.3	-0.7	(31)
(32)			MCD B	3.3	3.7	5.6	10.1	16.3	17.7	20.9	20.3	19.1	11.2	3.9	2.0	(32)
(33)		10%	WB	-1.0	-1.5	-0.1	2.5	7.3	10.0	12.6	12.4	9.5	4.3	-0.1	-2.5	(33)
(34)			MCD B	1.6	1.5	3.6	7.4	14.1	15.8	19.7	18.8	15.8	8.2	1.8	-0.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	14.6	14.9	13.8	14.4	14.2	15.3	15.5	15.0	12.7	12.3	13.0	(35)
(36)			MCD BR	13.1	16.1	16.9	18.2	21.5	20.4	21.4	21.3	22.4	18.1	13.9	14.2	(36)
(37)			MCWBR	9.3	11.4	11.1	10.5	11.2	10.3	10.7	11.1	12.0	10.8	9.9	10.5	(37)
(38)		5% WB	MCD BR	12.0	15.6	15.8	16.9	19.6	17.3	18.9	18.3	20.7	16.7	12.8	13.6	(38)
(39)			MCWBR	8.8	11.1	10.5	10.0	10.7	9.4	10.0	10.2	11.4	10.2	9.5	10.4	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.233	0.239	0.272	0.301	0.323	0.331	0.323	0.324	0.293	0.270	0.239	0.225	(40)	
(42)		taud	2.461	2.530	2.462	2.420	2.431	2.443	2.505	2.489	2.590	2.611	2.518	2.395	(41)	
(43)		Ebn at Noon	793	906	929	935	925	918	920	900	896	853	779	722	(42)	
		Edh at Noon	48	63	84	101	106	106	98	94	74	58	45	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.78	1.65	2.81	4.28	4.98	5.25	5.38	4.54	3.17	1.77	0.85	0.60	(44)	
(45)		RadStd	0.06	0.15	0.21	0.30	0.41	0.45	0.43	0.31	0.36	0.14	0.08	0.06	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page